A Additional Beyond-accuracy Results on the Yelp Dataset

Table 1: Performance with different sequential recommendation backbones

(a) BERT4Rec Backbone

	Embedding	Coverage	Gini	Serendipity	Novelty
			@10		
Beyond-accuracy	Trajectory-Only	0.08957±0.013	0.9657±0.0051	0.0377±0.0049	**13.26**±0.10
	Full Concatenation	0.02292±0.0046	0.9914±0.0013	0.04073±0.0042	12.23±0.37
	Modality-Weighted	0.04937±0.018	0.9825±0.0065	0.04716±0.0023	12.69±0.43
	FLAVA-Based	0.03321±0.018	0.9874±0.0065	0.04417±0.0019	12.49±0.43
	CoMM-Based	0.03319±0.013	0.9874±0.0049	0.04354±0.0015	12.76±0.20
	LUMIR_{MC}	0.1253±0.012	0.9516±0.0055	0.05196±0.0021	12.83±0.33
	$\text{LUMIR}_{\text{MC+SA}}$	**0.1515**±0.019	**0.9385**±0.010	**0.05739**±0.0057	12.88±0.48
			@20		
	Trajectory-Only	0.1431±0.019	0.9454±0.0074	0.06236±0.0058	**13.29**±0.10
	Full Concatenation	0.03882±0.0072	0.9845±0.0023	0.06191±0.0055	12.32±0.37
	Modality-Weighted	0.08263±0.029	0.9696±0.010	0.07205±0.0080	12.79±0.39
	FLAVA-Based	0.05606±0.029	0.9774±0.011	0.06434±0.0036	12.57±0.42
	CoMM-Based	0.05588±0.019	0.9776±0.0077	0.06571±0.0041	12.83±0.18
	LUMIR_{MC}	0.2004±0.017	0.9237±0.0080	0.08526±0.0054	12.90±0.29
	$\text{LUMIR}_{\text{MC+SA}}$	**0.2408**±0.025	**0.9057**±0.014	**0.09377**±0.0062	12.95±0.41

(b) SASRec Backbone

	Embedding	Coverage	Gini	Serendipity	Novelty
			@10		
Beyond-accuracy	Trajectory-Only	0.02308±0.013	0.9915±0.0035	0.02969±0.0051	11.58±0.65
	Full Concatenation	0.04858±0.0073	0.9833±0.0029	0.03277±0.0045	12.61±0.18
	Modality-Weighted	0.04860±0.011	0.9849±0.0042	0.04064±0.0028	12.52±0.21
	FLAVA-Based	0.04901±0.0097	0.9830±0.0035	0.04019±0.0025	12.23±0.14
	CoMM-Based	0.04633±0.016	0.9836±0.0048	0.03313±0.0059	12.52±0.47
	LUMIR_{MC}	0.1013±0.010	0.9658±0.0052	**0.05114**±0.0038	**12.73**±0.14
	$\text{LUMIR}_{\text{MC+SA}}$	**0.1434**±0.010	**0.9467**±0.0050	0.04327±0.0062	12.59±0.45
			@20		
	Trajectory-Only	0.03916±0.022	0.9847±0.0062	0.04779±0.0084	11.76±0.60
	Full Concatenation	0.07992±0.011	0.9719±0.0042	0.05413±0.0071	12.70±0.16
	Modality-Weighted	0.08107±0.018	0.9746±0.0068	0.06119±0.0059	12.65±0.21
	FLAVA-Based	0.08042±0.016	0.9717±0.0055	0.05947±0.0045	12.39±0.12
	CoMM-Based	0.07621±0.025	0.9719±0.0076	0.05367±0.0066	12.63±0.43
	LUMIR_{MC}	0.1607±0.014	0.9470±0.0072	**0.07974**±0.0041	**12.81**±0.15
	$\text{LUMIR}_{\text{MC+SA}}$	**0.2315**±0.018	**0.9161**±0.0076	0.06942±0.0083	12.74±0.39

(c) GRU4Rec Backbone

	Embedding	Coverage	Gini	Serendipity	Novelty
			@10		
Beyond-accuracy	Trajectory-Only	**0.06417**±0.013	**0.9826**±0.0031	0.03050±0.0039	11.62±0.11
	Full Concatenation	0.01215±0.0047	0.9949±0.0016	0.02552±0.0016	11.34±0.21
	Modality-Weighted	0.04122±0.012	0.9848±0.0046	0.03358±0.0057	11.99±0.29
	FLAVA-Based	0.02857±0.0089	0.9889±0.0035	0.03394±0.0046	11.77±0.24
	CoMM-Based	0.02786±0.0057	0.9893±0.0021	0.03114±0.0036	11.78±0.14
	LUMIR_{MC}	0.04398±0.019	0.9833±0.0067	**0.03892**±0.0045	**12.14**±0.44
	$\text{LUMIR}_{\text{MC+SA}}$	0.02653±0.011	0.9898±0.0039	0.03105±0.0045	11.73±0.38
			@20		
	Trajectory-Only	**0.1081**±0.027	0.9716±0.0048	0.04499±0.0040	11.83±0.095
	Full Concatenation	0.02090±0.0080	0.9905±0.0027	0.03856±0.0029	11.55±0.18
	Modality-Weighted	0.06947±0.021	0.9735±0.0077	**0.05621**±0.0060	12.16±0.27
	FLAVA-Based	0.04804±0.015	0.9804±0.0059	0.05377±0.0050	11.95±0.22
	CoMM-Based	0.04705±0.0093	0.9809±0.0034	0.04915±0.0044	11.96±0.12
	LUMIR_{MC}	0.07264±0.030	**0.9715**±0.011	0.05567±0.0047	**12.28**±0.39
	$\text{LUMIR}_{\text{MC+SA}}$	0.04502±0.018	0.9821±0.0065	0.04526±0.0064	11.91±0.35

Bold and underline indicate the highest mean value for each metric and the best-performing baseline, respectively. Results are averaged over 8 runs after removing the highest and lowest from 10 independent runs.